Questions by topic: Agroecosystems: abiotic & biotic factors - Set 1 (50 marks)

Answer all questions in the spaces provided.

0 1 Table 1 shows climate data for two sites where a farmer is considering growing maize, a frost-sensitive crop that needs a thermal growing season of at least 120 days.

Table 1

Feature	Site A: valley floor	Site B: south-facing slope
Frost-free period / days	110	150
Days with mean temperature above 5 °C	160	175
Mean summer rainfall / mm	180	150
Gradient / %	1	8

0 1 . 1 Use Table 1 to explain which site is more suitable for maize.

[3 marks]

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0 1 . 2 Describe **three** ways in which temperature can be controlled to increase crop productivity.

[3 marks]

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0 1 . 3 Explain why keeping livestock warm increases the amount of food converted to growth.

[2 marks]

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0 1 . 4 Explain how day length affects livestock production and how it can be manipulated.

[2 marks]

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0 2 . 1 Explain why a moderate water shortage reduces crop productivity even though the plants survive.

[3 marks]

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0 2 . 2 Explain the problems caused by waterlogged soils and describe **two** ways farmers reduce soil water levels.

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0 2 . 3 Explain how irrigating with groundwater can cause soil salinisation and how this harms crops.

[2 marks]

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0 2 . 4 State how a farmer could raise the pH of an acidic soil and explain why a low pH reduces fertility.

[1 mark]

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0 3 Table 1 compares organic and inorganic fertilisers.

0 3 . 1 Complete Table 1 by giving two advantages and two disadvantages of each type of fertiliser.

[6 marks]

Table 1

Fertiliser	Advantages	Disadvantages
Organic (manure, sewage sludge, compost)		
Inorganic (eg ammonium nitrate, NPK)		

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0 3 . 2 Explain how hydroponic cultivation maximises productivity and give **two** disadvantages.

[4 marks]

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0 4 Table 1 shows the yield of a vegetable crop under different pest control methods over three years on one farm.

Table 1

Method	Mean yield / t ha^{-1}	Pesticide applications per year	Non-target insect species recorded in field margins
Chemical insecticide only	18.5	6	8
Crop rotation + beetle banks + hedgerows	16.2	0	31
Integrated control	18.1	1-2	26

0 4 . 1 Explain how crop rotation and beetle banks reduce pest damage.

[4 marks]

0 4 . 2 Use Table 1 to evaluate integrated control compared with the other two methods.

[3 marks]

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0 4 . 3 Describe how the sterile male technique controls insect pests and explain why it is species-specific.

[3 marks]

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0 5 . 1 Describe **two** case studies of biological control, one successful and one that caused problems.

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0 5 . 2 Explain how pheromone traps can be used in pest control.

[2 marks]

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0 5 . 3 Explain why the use of antibiotics as growth promoters in livestock is a concern for human health.

[3 marks]

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0 5 . 4 Give **one** way farmers can support crop pollinators.

[1 mark]

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END OF QUESTIONS